TECHNISCHE UNIVERSITÄT MÜNCHEN

Quantum Chemistry Laboratory

Prof. N. Rösch

Full Solution of Exercise 15

Optimized geometry of F_d-H_d---F_a-H_a in comparison to experiment

		Becke3LYP	SVWN5	experiment
distance	F-F	2.747	2.569	2.72 ± 0.03
	F_d-H_d	0.929	0.946	
	F_a-H_a	0.925	0.937	
angle	F_a-F_d-H_d	8	10	10 ± 6
	F_d-F_a-H_a	116	107	117 ± 6
dihedral angle	F_d-H_d-F_a-H_a	0	0	

Distances in Å, angles in degrees, basis set 6-311++G(d,p).

Dissociation energy D_0 (0 K) in kJ/mol

Becke3LYP/6-311++G(d,p)	13.8

Becke3LYP/6-311++G(d,p) with counterpoise correction	11.8
SVWN5/6-311++G(d,p)	28.2
SVWN5/6-311++G(d,p) with counterpoise correction	25.7
experiment	12.7

[General help] [Overview] [Compute] [Contact the course supervisor]

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